

# Supervised Learning for Numerical Data

100 Questions: Linear Regression, Logistic Regression, LDA

GARP RAI Exam Preparation

# Q1: Simple Linear Regression Definition - Tutorial

**QUESTION:** What is simple linear regression?

**OPTIONS:**

- ❶ A model with multiple features and one target
- ❷ A model with one feature and one continuous target variable
- ❸ A model with no features
- ❹ A model for classification problems

**ANSWER: B**

**DETAILED EXPLANATION:**

- Simple linear regression is also called bivariate regression
- It involves just two variables: one target ( $y$ ) and one feature ( $x$ )
- Models how  $y$  changes in response to changes in  $x$
- Example: Stock return ( $y$ ) influenced by market return ( $x$ )

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** This describes MULTIPLE linear regression
- **C:** A model with no features would just be the mean
- **D:** Linear regression is for continuous targets, not classification

**EXAM TRAP:** Simple linear regression = one feature + one continuous target!

## Q2: Linear Regression Equation - Tutorial

**QUESTION:** What is the equation for a simple linear regression model?

**OPTIONS:**

- ❶  $y_i = \beta_0 + \beta_1 x_i + u_i$
- ❷  $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i$
- ❸  $y_i = \beta_0 + u_i$
- ❹  $y_i = \beta_1 x_i + u_i$

**ANSWER: A**

**DETAILED EXPLANATION:**

- $y_i$ : Target variable (dependent variable)
- $x_i$ : Feature (independent variable)
- $\beta_0$ : Intercept (bias) - value of  $y$  when  $x=0$
- $\beta_1$ : Slope (weight) - change in  $y$  for 1-unit change in  $x$
- $u_i$ : Error term - captures unexplained variation

**WHY OTHER OPTIONS ARE WRONG:**

- **B:** This is MULTIPLE linear regression
- **C:** Missing the feature term
- **D:** Missing the intercept term

**EXAM TRAP:** Simple linear regression has **intercept + slope + error!**

## Q3: Error Term Definition - Tutorial

**QUESTION:** What does the error term  $u_i$  represent in a regression model?

**OPTIONS:**

- ❶ The predicted value
- ❷ The intercept of the model
- ❸ Unexplained factors affecting  $y$  not in the model
- ❹ The slope of the model

**ANSWER: C**

**DETAILED EXPLANATION:**

- Captures all other factors affecting  $y$
- Includes inherent randomness
- Assumed to have mean of zero
- Assumed to have constant variance
- Assumed to be independent

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Predicted value is  $\hat{y}_i$
- **B:** Intercept is  $\beta_0$
- **D:** Slope is  $\beta_1$

**EXAM TRAP:** Error term captures **unexplained variation**!

## Q4: Parameter Estimation Methods - Tutorial

**QUESTION:** What are the three primary methods for estimating regression parameters?

**OPTIONS:**

- ① Least Squares, Maximum Likelihood, Method of Moments
- ② Gradient Descent, Backpropagation, Cross-Validation
- ③ LASSO, Ridge, Elastic Net
- ④ Linear, Logistic, Polynomial

**ANSWER: A**

**DETAILED EXPLANATION:**

- **Least Squares:** Most common (OLS)
- **Maximum Likelihood:** More general method
- **Method of Moments:** Another statistical technique
- OLS is most widely used for linear regression

**WHY OTHER OPTIONS ARE WRONG:**

- **B:** These are optimization/ML techniques
- **C:** These are regularization methods
- **D:** These are model types, not estimation methods

**EXAM TRAP:** Three estimation methods = OLS, MLE, Method of Moments!

**QUESTION: What is Ordinary Least Squares (OLS)?**

**OPTIONS:**

- ❶ A method that maximizes the likelihood of the data
- ❷ A method that minimizes the sum of squared differences between actual and predicted values
- ❸ A method that minimizes absolute errors
- ❹ A method that randomly selects parameters

**ANSWER: B**

**DETAILED EXPLANATION:**

- Most common estimation method for linear regression
- Finds  $\beta_0$  and  $\beta_1$  that minimize SSE
- $SSE = \sum (y_i - \hat{y}_i)^2$
- Squaring penalizes large errors more heavily

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** This describes Maximum Likelihood Estimation
- **C:** This describes Least Absolute Deviations (LAD)
- **D:** This describes random search

**EXAM TRAP:** OLS minimizes **sum of squared errors!**

# Q6: Multiple Linear Regression - Tutorial

**QUESTION:** What is multiple linear regression?

**OPTIONS:**

- ❶ A model with one feature and one target
- ❷ A model with multiple features and a continuous target
- ❸ A model for classification problems
- ❹ A model with no features

**ANSWER: B**

**DETAILED EXPLANATION:**

- Direct extension of simple linear regression
- Includes multiple features  $(x_1, x_2, \dots, x_m)$
- Equation:  $y_i = \beta_0 + \beta_1 x_{1i} + \dots + \beta_m x_{mi} + u_i$
- Each coefficient measures partial effect

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** This describes SIMPLE linear regression
- **C:** Multiple linear regression is for continuous targets
- **D:** A model with no features is just the mean

**EXAM TRAP:** Multiple linear regression = multiple features + continuous target!

## Q7: Multiple Linear Regression Equation - Tutorial

**QUESTION:** What is the equation for multiple linear regression with  $m$  features?

**OPTIONS:**

- ❶  $y_i = \beta_0 + \beta_1 x_i + u_i$
- ❷  $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_m x_{mi} + u_i$
- ❸  $y_i = \beta_0 + u_i$
- ❹  $y_i = \beta_1 x_{1i} + \beta_2 x_{2i} + u_i$

**ANSWER: B**

**DETAILED EXPLANATION:**

- $m$  features require  $m + 1$  parameters
- $\beta_0$ : intercept
- $\beta_1, \dots, \beta_m$ : slope parameters
- Each  $\beta$  measures partial effect
- OLS can still be used for estimation

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** This is SIMPLE linear regression
- **C:** Missing all feature terms
- **D:** Missing the intercept

**EXAM TRAP:** Multiple regression has  $m$  features + intercept =  $m+1$  parameters!

## Q8: Partial Effect Interpretation - Tutorial

**QUESTION:** In multiple regression, what does each  $\beta$  coefficient measure?

**OPTIONS:**

- ❶ The total effect of a feature
- ❷ The partial effect of a feature controlling for other features
- ❸ The correlation between features
- ❹ The error in the model

**ANSWER: B**

**DETAILED EXPLANATION:**

- Measures effect of one feature while holding others constant
- "Ceteris paribus" interpretation
- Controls for other variables in the model
- Example: Effect of experience on salary holding degree constant

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Total effect is different from partial effect
- **C:** Correlation is a different concept
- **D:** Error is  $u_i$ , not  $\beta$

**EXAM TRAP:**  $\beta$  measures **partial effect** controlling for other variables!

**QUESTION:** What is an interaction term in a regression model?

**OPTIONS:**

- ❶  $x_1 + x_2$  where features are added
- ❷  $x_1 \times x_2$  where features are multiplied
- ❸  $x_1^2$  where a feature is squared
- ❹  $\log(x_1)$  where a feature is transformed

**ANSWER: B**

**DETAILED EXPLANATION:**

- Equation:  $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{1i} x_{2i} + u_i$
- Captures complementarity between features
- Effect of  $x_1$  depends on  $x_2$  and vice versa
- Changes in both variables together impact  $y$

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** This is addition, not interaction
- **C:** This is a quadratic term
- **D:** This is a log transformation

**EXAM TRAP:** Interaction term = **features multiplied together!**

**QUESTION:** What does adding a squared term to a regression allow?

**OPTIONS:**

- ❶ A linear relationship
- ❷ A U-shaped or inverted U-shaped relationship
- ❸ No relationship between variables
- ❹ A perfectly flat relationship

**ANSWER: B**

**DETAILED EXPLANATION:**

- Equation:  $y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + u_i$
- Allows for nonlinear relationships
- U-shaped ( $\beta_2 > 0$ ): decreases then increases
- Inverted U-shaped ( $\beta_2 < 0$ ): increases then decreases
- Example: Experience and salary (diminishing returns)

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Linear relationship is without squared term
- **C:** There is still a relationship
- **D:** Flat relationship would require  $\beta_1 = \beta_2 = 0$

**EXAM TRAP:** Quadratic term allows **U-shaped relationships!**

# Q11: Omitting Relevant Features - Tutorial

**QUESTION:** What is the consequence of omitting a relevant feature from a regression?

**OPTIONS:**

- ❶ No consequence
- ❷ Biased parameter estimates for included features
- ❸ Smaller standard errors
- ❹ Better model fit

**ANSWER: B**

**DETAILED EXPLANATION:**

- Omitted variable bias
- Estimates are systematically wrong
- Bias does not disappear with larger samples
- Model attributes influence of missing variable to included ones

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** There are significant consequences
- **C:** Standard errors may be biased too
- **D:** Model fit is compromised

**EXAM TRAP:** Omitting relevant features causes **biased estimates!**

## Q12: Including Irrelevant Features - Tutorial

**QUESTION:** What is the consequence of including an irrelevant feature?

**OPTIONS:**

- ❶ Biased estimates for relevant features
- ❷ Inefficient (less precise) parameter estimates
- ❸ No consequence at all
- ❹ Smaller standard errors

**ANSWER: B**

**DETAILED EXPLANATION:**

- Typically not biased for relevant variables
- Causes inefficiency - larger standard errors
- Model becomes overly complex
- Risk of overfitting

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Relevant estimates are usually unbiased
- **C:** There is a consequence (inefficiency)
- **D:** Standard errors increase, not decrease

**EXAM TRAP:** Irrelevant features cause **inefficiency** (larger standard errors)!

## Q13: Incorrect Functional Form - Tutorial

**QUESTION:** What happens if we assume a linear relationship when the true relationship is non-linear?

**OPTIONS:**

- ❶ The model will fit perfectly
- ❷ The model will be a poor representation of reality
- ❸ The model will have no errors
- ❹ The model will be more efficient

**ANSWER: B**

**DETAILED EXPLANATION:**

- Model misspecification
- Biased predictions
- Incorrect inferences about feature impacts
- Example: Linear model for quadratic relationship

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Misspecified models don't fit perfectly
- **C:** There will be systematic errors
- **D:** Efficiency is compromised

**EXAM TRAP:** Wrong functional form = poor model representation!

**QUESTION:** What is multicollinearity in regression?

**OPTIONS:**

- ❶ When features are highly correlated with the target
- ❷ When features are highly correlated with each other
- ❸ When the error term has non-constant variance
- ❹ When there are outliers in the data

**ANSWER: B**

**DETAILED EXPLANATION:**

- Features (independent variables) are highly correlated
- Makes it hard to isolate individual effects
- Two types: perfect and imperfect
- Example: Square footage and number of bedrooms

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Correlation with target is desirable
- **C:** This is heteroskedasticity
- **D:** This is a different problem

**EXAM TRAP:** Multicollinearity = features correlated with each other!

# Q15: Perfect Multicollinearity - Tutorial

**QUESTION:** What happens with perfect multicollinearity?

**OPTIONS:**

- ❶ OLS estimation works normally
- ❷ OLS estimation breaks down; separate effects cannot be estimated
- ❸ Standard errors become smaller
- ❹ The model becomes more efficient

**ANSWER: B**

**DETAILED EXPLANATION:**

- Exact linear relationship between features
- Example: Celsius and Fahrenheit included together
- OLS cannot estimate separate coefficients
- Must remove one or more features

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** OLS fails with perfect multicollinearity
- **C:** Standard errors are not valid
- **D:** Model is not efficient

**EXAM TRAP:** Perfect multicollinearity = OLS breaks down!

**QUESTION:** What is imperfect multicollinearity?

**OPTIONS:**

- ❶ Features are exactly linearly related
- ❷ Features are very closely (but not perfectly) correlated
- ❸ Features are completely uncorrelated
- ❹ Features are correlated with the error term

**ANSWER: B**

**DETAILED EXPLANATION:**

- Correlation very high (e.g., 0.9 or -0.95)
- More common than perfect collinearity
- Causes large standard errors
- Makes it hard to separate effects

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** This is perfect multicollinearity
- **C:** This is no multicollinearity
- **D:** This is endogeneity, not multicollinearity

**EXAM TRAP:** Imperfect multicollinearity = **very high but not perfect correlation!**

**QUESTION: What is an outlier in regression?**

**OPTIONS:**

- ❶ A data point that fits the pattern perfectly
- ❷ An anomalous data point far from the bulk of observations
- ❸ The mean of the data
- ❹ The median of the data

**ANSWER: B**

**DETAILED EXPLANATION:**

- Data point with large residual
- Far from the regression line
- Can be due to data entry errors
- Or rare but genuine events

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** This is an inlier, not an outlier
- **C:** The mean is a summary statistic
- **D:** The median is a summary statistic

**EXAM TRAP:** Outliers are **anomalous data points far from others!**

**QUESTION:** Why can outliers have a large impact on OLS estimates?

**OPTIONS:**

- ❶ OLS ignores outliers
- ❷ OLS minimizes squared residuals, so large residuals have disproportionate influence
- ❸ OLS gives equal weight to all observations
- ❹ OLS uses absolute residuals

**ANSWER: B**

**DETAILED EXPLANATION:**

- Squaring residuals amplifies large differences
- Points far from line pull the line toward them
- Can significantly change parameter estimates
- Example: One machine with many failures

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** OLS does NOT ignore outliers
- **C:** Squaring gives more weight to large residuals
- **D:** OLS uses squared, not absolute residuals

**EXAM TRAP:** Outliers have **disproportionate influence** because of squared errors!

**QUESTION:** What does Cook's Distance measure?

**OPTIONS:**

- ❶ The mean of the residuals
- ❷ The effect of deleting an observation on all fitted values
- ❸ The correlation between features
- ❹ The variance of the error term

**ANSWER: B**

**DETAILED EXPLANATION:**

- Diagnostic for influential points
- Measures how much fitted values change when point is removed
- Large values indicate influential observations
- Higher than others warrants investigation

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Mean of residuals is different
- **C:** This is correlation
- **D:** This is error variance

**EXAM TRAP:** Cook's Distance identifies influential observations!

**QUESTION: What is heteroskedasticity in regression?**

**OPTIONS:**

- ❶ Constant variance of errors
- ❷ Non-constant variance of errors
- ❸ Zero mean of errors
- ❹ Independence of errors

**ANSWER: B**

**DETAILED EXPLANATION:**

- Error variance is not constant
- Opposite of homoskedasticity
- Common in cross-sectional data
- Example: Profits of small vs large companies

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** This is homoskedasticity
- **C:** This is an OLS assumption
- **D:** This is another OLS assumption

**EXAM TRAP:** Heteroskedasticity = non-constant error variance!

**QUESTION:** How can heteroskedasticity be detected?

**OPTIONS:**

- ① Looking at residual plots for changing spread
- ② Looking at the mean of residuals
- ③ Looking at the R-squared
- ④ Looking at the intercept

**ANSWER: A**

**DETAILED EXPLANATION:**

- Plot residuals vs fitted values
- Look for funnel shape
- Spread changes with fitted values
- Formal test: White's Test

**WHY OTHER OPTIONS ARE WRONG:**

- **B:** Mean of residuals is usually zero
- **C:** R-squared doesn't detect heteroskedasticity
- **D:** Intercept doesn't detect heteroskedasticity

**EXAM TRAP:** Heteroskedasticity = **funnel shape in residual plot!**

**QUESTION: What is White's Test used for?**

**OPTIONS:**

- ❶ Detecting multicollinearity
- ❷ Detecting heteroskedasticity
- ❸ Detecting outliers
- ❹ Detecting autocorrelation

**ANSWER: B**

**DETAILED EXPLANATION:**

- Formal statistical test for heteroskedasticity
- Standard option in most regression software
- Significant result indicates heteroskedasticity
- Produces White's heteroskedasticity-consistent errors

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Variance Inflation Factor (VIF) detects multicollinearity
- **C:** Cook's Distance detects outliers
- **D:** Durbin-Watson detects autocorrelation

**EXAM TRAP:** White's Test = [heteroskedasticity detection](#)!

**QUESTION:** What is one remedy for heteroskedasticity?

**OPTIONS:**

- ❶ Add more features
- ❷ Use Weighted Least Squares or robust standard errors
- ❸ Remove all features
- ❹ Ignore it

**ANSWER: B**

**DETAILED EXPLANATION:**

- **WLS:** Weight observations by inverse variance
- **Robust errors:** White's heteroskedasticity-consistent errors
- **Log transformation:** Of variables can help
- Select "robust errors" option in software

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Adding features doesn't fix heteroskedasticity
- **C:** Removing all features removes the model
- **D:** Ignoring it is not a solution

**EXAM TRAP:** Remedies = WLS, robust errors, or log transformation!

**QUESTION:** What is the purpose of stepwise regression?

**OPTIONS:**

- ❶ To estimate parameters
- ❷ To automate feature selection
- ❸ To create new features
- ❹ To transform variables

**ANSWER:** B

**DETAILED EXPLANATION:**

- Selects subset of features from candidate pool
- Reduces model complexity
- Avoids redundant features
- Uses AIC or other criteria

**WHY OTHER OPTIONS ARE WRONG:**

- A: Parameter estimation is OLS
- C: Feature creation is different
- D: Variable transformation is different

**EXAM TRAP:** Stepwise regression = automated feature selection!

## Q25: Forward vs Backward Selection - Tutorial

**QUESTION:** How does forward selection differ from backward selection?

**OPTIONS:**

- ❶ Forward starts with no features; backward starts with all features
- ❷ Forward starts with all features; backward starts with no features
- ❸ They are identical
- ❹ Forward removes features; backward adds features

**ANSWER: A**

**DETAILED EXPLANATION:**

- **Forward:** Start with intercept only, add features one by one
- **Backward:** Start with all features, remove one by one
- Both use AIC to decide
- May yield different final models

**WHY OTHER OPTIONS ARE WRONG:**

- **B:** This is reversed
- **C:** They are different approaches
- **D:** Forward adds; backward removes

**EXAM TRAP:** Forward = add features; Backward = remove features!

**QUESTION:** What is a classification problem in machine learning?

**OPTIONS:**

- ❶ Predicting a continuous numerical value
- ❷ Assigning observations to predefined categories
- ❸ Clustering similar observations
- ❹ Reducing dimensionality

**ANSWER: B**

**DETAILED EXPLANATION:**

- Target is categorical (not continuous)
- Binary: two categories (0/1)
- Multi-class: more than two categories
- Examples: Default/No Default, Fraud/Legitimate

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** This is regression
- **C:** This is unsupervised learning
- **D:** This is feature reduction

**EXAM TRAP:** Classification = [assigning to categories](#)!

**QUESTION:** Which is NOT an example of binary classification in risk management?

**OPTIONS:**

- ❶ Loan default (Yes/No)
- ❷ Fraud detection (Fraudulent/Legitimate)
- ❸ Predicting house prices
- ❹ Cybersecurity threat (Malicious/Benign)

**ANSWER: C**

**DETAILED EXPLANATION:**

- Binary classification has two categories
- Predicting house prices is regression (continuous)
- Default prediction, fraud detection, threat detection are all binary

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Yes/No default is binary classification
- **B:** Fraudulent/Legitimate is binary classification
- **D:** Malicious/Benign is binary classification

**EXAM TRAP:** Predicting prices = regression, not classification!

**QUESTION:** Why is linear regression inappropriate for binary classification?

**OPTIONS:**

- ❶ Linear regression is too complex
- ❷ Predictions can fall outside the 0-1 range
- ❸ Linear regression requires categorical features
- ❹ Linear regression is always accurate

**ANSWER: B**

**DETAILED EXPLANATION:**

- Linear regression can predict  $< 0$  or  $> 1$
- No constraint to keep predictions between 0 and 1
- 0 and 1 are not probabilities
- Logistic regression solves this with sigmoid function

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Linear regression is actually simpler
- **C:** Linear regression doesn't require categorical features
- **D:** Linear regression is NOT accurate for binary outcomes

**EXAM TRAP:** Linear regression can give probabilities outside 0-1 range!

**QUESTION:** What is the logistic function also called?

**OPTIONS:**

- 1 ReLU function
- 2 Sigmoid function
- 3 Tanh function
- 4 Linear function

**ANSWER: B**

**DETAILED EXPLANATION:**

- Logistic function = Sigmoid function
- Bounded between 0 and 1
- S-shaped curve
- Formula:  $F(y) = \frac{1}{1+e^{-y}}$

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** ReLU is a different activation function
- **C:** Tanh ranges from -1 to 1
- **D:** Linear function is unbounded

**EXAM TRAP:** Logistic = Sigmoid function!

## Q30: Logistic Regression Equation - Tutorial

**QUESTION:** What is the logistic regression model formula for  $P(y_i = 1)$ ?

**OPTIONS:**

❶  $P_i = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_{1i} + \dots + \beta_m x_{mi})}}$

❷  $P_i = \beta_0 + \beta_1 x_{1i} + \dots + \beta_m x_{mi}$

❸  $P_i = e^{(\beta_0 + \beta_1 x_{1i} + \dots + \beta_m x_{mi})}$

❹  $P_i = \frac{1}{1 + e^{(\beta_0 + \beta_1 x_{1i} + \dots + \beta_m x_{mi})}}$

**ANSWER: A**

**DETAILED EXPLANATION:**

- Logistic function applied to linear combination
- Bounds probability between 0 and 1
- $\beta_0 + \beta_1 x_{1i} + \dots + \beta_m x_{mi}$  is the linear predictor
- $e$  is Euler's number ( $\approx 2.718$ )

**WHY OTHER OPTIONS ARE WRONG:**

- **B:** This is linear regression, not logistic
- **C:** This is exponential, not logistic
- **D:** Wrong sign in the exponent (should be negative)

**EXAM TRAP:** Logistic regression =  $1/(1 + e^{-\text{linear\_predictor}})$ !

**QUESTION: What does logistic regression output?**

**OPTIONS:**

- ❶ A continuous value with no bounds
- ❷ A probability between 0 and 1
- ❸ Either 0 or 1 only
- ❹ A negative number

**ANSWER: B**

**DETAILED EXPLANATION:**

- Outputs probability  $P(y_i = 1)$
- Always between 0 and 1
- Can be thresholded to get binary prediction
- Typically threshold = 0.5

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Linear regression gives unbounded output
- **C:** Output is probability, thresholding gives 0/1
- **D:** Probabilities are never negative

**EXAM TRAP:** Logistic regression outputs **probabilities between 0 and 1!**

**QUESTION:** How are logistic regression coefficients interpreted?

**OPTIONS:**

- ❶ As linear effects on the probability
- ❷ As linear effects on the log-odds
- ❸ As linear effects on the target directly
- ❹ Coefficients cannot be interpreted

**ANSWER: B**

**DETAILED EXPLANATION:**

- Logistic transformation is nonlinear
- Coefficients affect log-odds linearly
- $\text{Log-odds} = \log\left(\frac{p}{1-p}\right)$
- Signs and significance are still informative

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Effect on probability is nonlinear
- **C:** Target is binary, not continuous
- **D:** Coefficients CAN be interpreted (log-odds)

**EXAM TRAP:** Logistic coefficients affect **log-odds**, not probability directly!

**QUESTION:** In the LendingClub example, which factors significantly affect default probability?

**OPTIONS:**

- ① Amount and Income
- ② Term and Interest Rate
- ③ Homeowner and Delinquent
- ④ Bankruptcies and Employment History

**ANSWER: B**

**DETAILED EXPLANATION:**

- Term: longer loans increase default probability
- Interest rate: higher rates increase default probability
- Both significant at 5
- Mortgage has negative effect (significant)

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Amount and Income are not significant
- **C:** Homeowner and Delinquent are not significant
- **D:** Bankruptcies and Employment are not significant

**EXAM TRAP:** Term and Interest Rate = **significant predictors** of default!

**QUESTION:** In the LendingClub example, what does a negative coefficient on "Mortgage" mean?

**OPTIONS:**

- ❶ Having a mortgage increases default probability
- ❷ Having a mortgage decreases default probability
- ❸ Mortgage has no effect on default
- ❹ Mortgage is the most important factor

**ANSWER: B**

**DETAILED EXPLANATION:**

- Negative coefficient = lower default probability
- Mortgage holders are less likely to default
- Coefficient is significant at 5
- May indicate financial stability

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Negative means decrease, not increase
- **C:** It is significant, so it has an effect
- **D:** Other factors may be more important

**EXAM TRAP:** Negative coefficient = decrease in default probability!

**QUESTION:** What does a coefficient with denote in regression output?

**OPTIONS:**

- ❶ The coefficient is zero
- ❷ The coefficient is significant at 1
- ❸ The coefficient is significant at 5
- ❹ The coefficient is not significant

**ANSWER: C**

**DETAILED EXPLANATION:**

- significant at 5% level
- significant at 1% level
- significant at 10% level
- Indicates probability that coefficient is zero is low

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** two means not zero (significant)
- **B:** 3 is 1% significance
- **D:** 2 means it IS significant

**EXAM TRAP:** 2 = 5% significance; 3 tar = 1% significance!

**QUESTION:** What was the outcome of the Global Financial Company's logistic regression application?

**OPTIONS:**

- ❶ 5% increase in business
- ❷ 9.8% increase in business
- ❸ 15% increase in business
- ❹ No change in business

**ANSWER: B**

**DETAILED EXPLANATION:**

- Used logistic regression to refine lending offers
- Predicted likelihood of clients accepting loan proposals
- Achieved 9.8% increase in business
- Enabled more targeted and effective offers

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Lower than actual increase
- **C:** Higher than actual increase
- **D:** There was a significant increase

**EXAM TRAP:** Global Financial Company achieved 9.8% business increase!

**QUESTION:** What is multinomial logit used for?

**OPTIONS:**

- ① Binary classification
- ② Multi-class classification with unordered categories
- ③ Multi-class classification with ordered categories
- ④ Continuous prediction

**ANSWER: B**

**DETAILED EXPLANATION:**

- Extends logistic regression to  $\geq 2$  categories
- Categories are nominal (no ordering)
- One category is the baseline/reference
- Estimates log-odds relative to baseline

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Binary is standard logistic regression
- **C:** Ordered logit is for ordered categories
- **D:** Regression is for continuous prediction

**EXAM TRAP:** Multinomial logit = **unordered multi-class classification!**

**QUESTION:** Which is an example of a discrete choice problem?

**OPTIONS:**

- ❶ Predicting house prices
- ❷ Commuter choosing between car, bus, bicycle, walking
- ❸ Predicting stock returns
- ❹ Estimating company revenue

**ANSWER: B**

**DETAILED EXPLANATION:**

- Multiple unordered categories
- No inherent ranking between options
- Features: age, income, distance, fitness
- Multinomial logit is appropriate

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** House price prediction is regression
- **C:** Stock return prediction is regression
- **D:** Revenue estimation is regression

**EXAM TRAP:** Discrete choice = choosing among unordered categories!

**QUESTION:** What is ordered logit used for?

**OPTIONS:**

- ❶ Binary classification
- ❷ Multi-class with ordered categories
- ❸ Unordered multi-class classification
- ❹ Continuous prediction

**ANSWER: B**

**DETAILED EXPLANATION:**

- Categories have natural ordering
- Examples: Credit ratings (AAA, AA, A, BBB...)
- Risk scores (Low, Medium, High)
- Uses threshold/cutoff parameters

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Binary is standard logistic regression
- **C:** Multinomial logit is for unordered
- **D:** Regression is for continuous prediction

**EXAM TRAP:** Ordered logit = ordered multi-class classification!

**QUESTION:** Which is an example of ordinal data?

**OPTIONS:**

- ❶ Loan default (Yes/No)
- ❷ Credit ratings (AAA, AA, A, BBB)
- ❸ Commuter transport mode choice
- ❹ House prices

**ANSWER: B**

**DETAILED EXPLANATION:**

- Ordinal data has natural ordering
- AAA is better than AA, AA better than A
- Ordered logit is appropriate
- Other examples: survey responses, risk scores

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Binary (no ordering)
- **C:** Nominal (no ordering)
- **D:** Continuous (not categorical)

**EXAM TRAP:** Credit ratings = ordinal data with natural ordering!

**QUESTION:** What does a log-odds of zero indicate?

**OPTIONS:**

- ① The category is infinitely more likely
- ② The category is equally likely as the baseline
- ③ The category is impossible
- ④ The category is certain

**ANSWER: B**

**DETAILED EXPLANATION:**

- Log-odds =  $\log\left(\frac{P}{1-P}\right)$
- Zero means  $\frac{P}{1-P} = 1$
- So  $P = 0.5$
- Equal probability of both outcomes

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Positive means more likely
- **C:** Zero means equal, not impossible
- **D:** Zero means equal, not certain

**EXAM TRAP:** Log-odds of 0 = equal probability!

**QUESTION:** What is the prior probability in LDA?

**OPTIONS:**

- ❶ Probability of class before seeing data
- ❷ Probability of class after seeing data
- ❸ The feature distribution
- ❹ The error term

**ANSWER: A**

**DETAILED EXPLANATION:**

- $\pi_j$  in LDA formula
- Unconditional probability of each class
- Can be assumed equal
- Or estimated from training data frequencies

**WHY OTHER OPTIONS ARE WRONG:**

- **B:** This is posterior probability
- **C:** This is likelihood
- **D:** This is different concept

**EXAM TRAP:** Prior = probability before seeing data!

## Q43: Posterior Probability in LDA - Tutorial

**QUESTION:** How is a new observation classified in LDA?

**OPTIONS:**

- ❶ Randomly assigned
- ❷ Assigned to class with highest discriminant score
- ❸ Assigned to class with lowest discriminant score
- ❹ Assigned to the majority class

**ANSWER: B**

**DETAILED EXPLANATION:**

- Calculate  $\delta_j(x)$  for each class
- Choose class with largest  $\delta_j(x)$
- This maximizes posterior probability
- Discriminant score is linear function of  $x$

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Classification is deterministic
- **C:** Highest score gives best class
- **D:** Uses discriminants, not just majority

**EXAM TRAP:** LDA assigns to class with highest discriminant score!

**QUESTION:** What are the key assumptions of LDA?

**OPTIONS:**

- ❶ Features are independent and have equal means
- ❷ Features are normally distributed with common covariance matrix
- ❸ Features are uniformly distributed
- ❹ Features are categorical

**ANSWER:** B

**DETAILED EXPLANATION:**

- Multivariate normal distribution for each class
- Common covariance matrix across classes
- Features are continuous
- Different mean vectors for each class

**WHY OTHER OPTIONS ARE WRONG:**

- A: Means are different, not equal
- C: Features are normal, not uniform
- D: Features are continuous, not categorical

**EXAM TRAP:** LDA = normal distributions + common covariance!

**QUESTION:** What happens if LDA assumptions are not perfectly met?

**OPTIONS:**

- ❶ LDA cannot be used at all
- ❷ LDA often remains robust and performs well
- ❸ LDA always fails completely
- ❹ LDA requires no assumptions

**ANSWER: B**

**DETAILED EXPLANATION:**

- Assumptions are ideal conditions
- In practice, often not perfectly met
- LDA is quite robust to violations
- Often performs well in practice

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** LDA can still be used
- **C:** LDA often works well
- **D:** LDA does have assumptions

**EXAM TRAP:** LDA is **robust** even when assumptions are violated!

**QUESTION:** What is the primary purpose of Linear Discriminant Analysis (LDA)?

**OPTIONS:**

- ❶ Regression analysis
- ❷ Classification with multiple well-separated classes
- ❸ Feature engineering
- ❹ Error correction

**ANSWER: B**

**DETAILED EXPLANATION:**

- Finds linear combinations that best separate classes
- Maximizes between-class separation
- Minimizes within-class variance
- Alternative to logistic regression for multi-class

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** LDA is for classification, not regression
- **C:** LDA is not for feature engineering
- **D:** LDA is not for error correction

**EXAM TRAP:** LDA is for **classification with multiple classes!**

**QUESTION:** What is a key advantage of LDA over logistic regression?

**OPTIONS:**

- ❶ LDA is always more accurate
- ❷ LDA handles multi-class problems naturally
- ❸ LDA requires no assumptions
- ❹ LDA is faster to compute

**ANSWER: B**

**DETAILED EXPLANATION:**

- Logistic regression is naturally binary
- LDA extends directly to multiple classes
- Multi-class logistic regression requires extensions
- LDA gives linear discriminant functions

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Neither is always more accurate
- **C:** LDA has assumptions (normality, common covariance)
- **D:** Speed depends on implementation

**EXAM TRAP:** LDA = natural multi-class classification!

## Q48: LDA Discriminant Function - Tutorial

**QUESTION:** What is the LDA discriminant function formula?

**OPTIONS:**

①  $\delta_j(x) = x^T \Sigma^{-1} \mu_j - \frac{1}{2} \mu_j^T \Sigma^{-1} \mu_j + \log(\pi_j)$

②  $\delta_j(x) = x^T \Sigma \mu_j$

③  $\delta_j(x) = \mu_j^T \Sigma^{-1} x$

④  $\delta_j(x) = \log(\pi_j) + \frac{1}{2} \mu_j^T \Sigma^{-1} \mu_j$

**ANSWER: A**

**DETAILED EXPLANATION:**

- $x$ : feature vector
- $\mu_j$ : mean vector for class  $j$
- $\Sigma$ : common covariance matrix
- $\pi_j$ : prior probability of class  $j$
- First term:  $x$ 's contribution, linear in  $x$

**WHY OTHER OPTIONS ARE WRONG:**

- **B:** Missing important terms
- **C:** Missing important terms
- **D:** Missing the  $x$  term

**EXAM TRAP:** LDA discriminant function = linear in  $x$  with prior adjustment!

**QUESTION:** In the LDA example, what are the prior probabilities for default/no-default?

**OPTIONS:**

- ☒ 1 0.5 and 0.5
- ☐ 2 0.7 and 0.3
- ☐ 3 0.8 and 0.2
- ☐ 4 1.0 and 0.0

**ANSWER: B**

**DETAILED EXPLANATION:**

- 7 out of 10 are no default (70)
- 3 out of 10 are default (30)
- Priors based on class frequencies
- $\pi_0 = 0.7$  (no default),  $\pi_1 = 0.3$  (default)

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Not balanced (7 vs 3)
- **C:** Not 8 vs 2
- **D:** Not all one class

**EXAM TRAP:** Priors = class frequencies in training data (0.7, 0.3)!

**QUESTION:** In the LDA example, what is the mean balance for non-defaults?

**OPTIONS:**

- 1 1.31
- 2 -0.56
- 3 -0.74
- 4 0.32

**ANSWER: B**

**DETAILED EXPLANATION:**

- $\mu_0 = \begin{pmatrix} -0.56 \\ 0.32 \end{pmatrix}$
- First element is mean balance for non-defaults
- Second element is mean income for non-defaults

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** This is mean balance for defaults
- **C:** This is mean income for defaults
- **D:** This is mean income for non-defaults

**EXAM TRAP:** Non-default balance mean = -0.56 (standardized)!

**QUESTION:** In the LDA example, how is the new borrower classified?

**OPTIONS:**

- 1 Default
- 2 No default
- 3 Cannot be classified
- 4 Both classes equally likely

**ANSWER: B**

**DETAILED EXPLANATION:**

- $\delta_0 = 0.30$  (no default)
- $\delta_1 = -3.95$  (default)
- Since  $\delta_0 > \delta_1$ , classify as no default
- Discriminant score is much higher for no default

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Default score is lower
- **C:** We have enough information
- **D:** Scores are not equal (0.30 vs -3.95)

**EXAM TRAP:** Classify as **no default** (higher discriminant score)!

**QUESTION:** What does the covariance matrix in LDA measure?

**OPTIONS:**

- ❶ The means of features
- ❷ The spread and correlation structure of features
- ❸ The class probabilities
- ❹ The error term

**ANSWER: B**

**DETAILED EXPLANATION:**

- Variances on the diagonal
- Covariances off-diagonal
- Assumed to be common across classes
- $\Sigma$  in the discriminant function

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Means are  $\mu_j$
- **C:** Probabilities are  $\pi_j$
- **D:** Error is different concept

**EXAM TRAP:** Covariance matrix = spread and correlation of features!

**QUESTION:** Why are the features standardized in the LDA example?

**OPTIONS:**

- ❶ To make them categorical
- ❷ To ensure equal scale for LDA
- ❸ To remove outliers
- ❹ To make the data smaller

**ANSWER: B**

**DETAILED EXPLANATION:**

- LDA assumes continuous features
- Features may have different scales
- Standardization gives mean 0, variance 1
- Ensures features are comparable

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Standardization doesn't make categorical
- **C:** Standardization doesn't remove outliers
- **D:** Standardization doesn't change data size

**EXAM TRAP:** Standardization = **equal scales** for LDA!

**QUESTION:** What is another use of LDA beyond classification?

**OPTIONS:**

- ❶ Feature engineering
- ❷ Dimensionality reduction
- ❸ Error detection
- ❹ Data cleaning

**ANSWER: B**

**DETAILED EXPLANATION:**

- Projects data to lower-dimensional space
- Maximizes between-class separation
- Similar to PCA but supervised
- Can reduce to  $K - 1$  dimensions

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** LDA uses features, doesn't engineer them
- **C:** LDA is not for error detection
- **D:** LDA is not for data cleaning

**EXAM TRAP:** LDA can be used for **dimensionality reduction**!

**QUESTION:** How does LDA differ from PCA for dimensionality reduction?

**OPTIONS:**

- ☒ 1 LDA is unsupervised; PCA is supervised
- ☐ 2 LDA is supervised; PCA is unsupervised
- ☐ 3 Both are supervised
- ☐ 4 Both are unsupervised

**ANSWER: B**

**DETAILED EXPLANATION:**

- ☐ LDA uses class labels (supervised)
- ☐ PCA ignores class labels (unsupervised)
- ☐ LDA maximizes class separation
- ☐ PCA maximizes variance

**WHY OTHER OPTIONS ARE WRONG:**

- ☐ A: LDA is supervised, not unsupervised
- ☐ C: PCA is unsupervised
- ☐ D: LDA is supervised

**EXAM TRAP:** LDA = supervised; PCA = unsupervised!

## Q56: Salary Prediction Example - Tutorial

**QUESTION:** In the salary example, what does the intercept of 16.88 mean?

**OPTIONS:**

- ❶ Salary increases by \$16.88 per year
- ❷ Expected salary with no experience is \$16.88/hr
- ❸ Maximum possible salary is \$16.88/hr
- ❹ Salary decreases by \$16.88 per year

**ANSWER: B**

**DETAILED EXPLANATION:**

- Intercept  $\beta_0 = 16.88$
- Expected salary when experience = 0
- "Someone just joining with no experience"
- In dollars per hour

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** This is the slope (1.06)
- **C:** Not the maximum possible salary
- **D:** Salary increases, doesn't decrease

**EXAM TRAP:** Intercept = expected value when feature = 0!

**QUESTION:** In the salary example, what does the slope of 1.06 mean?

**OPTIONS:**

- ❶ Each year of experience increases salary by \$1.06/hr
- ❷ Each year of experience decreases salary by 1.06/hr
- ❸ Salary is \$1.06/hr with no experience
- ❹ Maximum salary is \$1.06/hr

**ANSWER: A**

**DETAILED EXPLANATION:**

- Slope  $\beta_1 = 1.06$
- For each additional year of experience
- Salary increases by \$1.06 per hour
- Linear relationship

**WHY OTHER OPTIONS ARE WRONG:**

- **B:** Slope is positive, so increases
- **C:** This is the intercept (16.88)
- **D:** Not a maximum

**EXAM TRAP:** Slope = change in y for 1-unit change in x!

**QUESTION:** Why was a quadratic term added to the salary model?

**OPTIONS:**

- ❶ To make the model linear
- ❷ To capture a non-linear relationship between experience and salary
- ❸ To add more features
- ❹ To reduce model complexity

**ANSWER: B**

**DETAILED EXPLANATION:**

- Visual inspection suggested non-linearity
- Salary increases quickly then plateaus
- Quadratic term allows U-shaped relationship
- Better fit to the data

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Quadratic makes it non-linear
- **C:** It adds a feature but that's not the reason
- **D:** It increases complexity

**EXAM TRAP:** Quadratic term captures **non-linear relationships!**

**QUESTION:** At what experience level is salary maximized in the quadratic model?

**OPTIONS:**

- ☒ 1 10.3 years
- ☐ 2 17.3 years
- ☐ 3 25.0 years
- ☐ 4 5.0 years

**ANSWER: B**

**DETAILED EXPLANATION:**

- Model:  $\hat{y} = 11.82 + 2.77x - 0.08x^2$
- Derivative:  $\frac{dy}{dx} = 2.77 - 0.16x$
- Set to zero:  $2.77 - 0.16x = 0$
- $x = 2.77/0.16 = 17.31$  years

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Too low
- **C:** Too high
- **D:** Too low

**EXAM TRAP:** Maximization occurs at 17.3 years of experience!

**QUESTION:** In the salary model with degree dummy, what does 0.92 represent?

**OPTIONS:**

- 1 Salary increase of \$0.92/hr for having a degree
- 2 Salary decrease of \$0.92/hr for having a degree
- 3 Experience effect
- 4 Intercept

**ANSWER: A**

**DETAILED EXPLANATION:**

- Coefficient on degree dummy = 0.92
- Degree holders earn \$0.92/hr more
- Holding experience constant
- "Compared with someone having identical experience but no degree"

**WHY OTHER OPTIONS ARE WRONG:**

- B: Positive means increase
- C: This is the degree effect
- D: Intercept is 11.38

**EXAM TRAP:** Dummy coefficient = *effect of category* compared to baseline!

**QUESTION: What is a dummy variable?**

**OPTIONS:**

- ❶ A continuous variable
- ❷ A binary variable (0 or 1) representing categories
- ❸ A variable with more than two categories
- ❹ A variable that is always zero

**ANSWER: B**

**DETAILED EXPLANATION:**

- Takes values 0 or 1
- Represents categorical data
- Example: Degree (1 = has degree, 0 = no degree)
- Can be used in regression models

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Dummy variables are binary
- **C:** Multiple categories need multiple dummies
- **D:** Dummy variables can be 0 or 1

**EXAM TRAP:** Dummy variables are **binary (0/1)**!

**QUESTION:** What is the dummy variable trap?

**OPTIONS:**

- ❶ Including too many dummy variables
- ❷ Including a set of dummies that sum to one along with an intercept
- ❸ Not including enough dummy variables
- ❹ Including only one dummy variable

**ANSWER: B**

**DETAILED EXPLANATION:**

- Example: Gender dummies (male, female) + intercept
- Dummies sum to 1 for every observation
- Perfect multicollinearity
- OLS breaks down

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Too many is not the trap
- **C:** Not enough is not the trap
- **D:** One dummy is fine

**EXAM TRAP:** Dummy variable trap = perfect multicollinearity with intercept!

**QUESTION:** What are seasonal dummies used for in time-series regression?

**OPTIONS:**

- ❶ To remove seasonality
- ❷ To capture patterns that repeat weekly, monthly, or quarterly
- ❸ To make data stationary
- ❹ To remove trends

**ANSWER: B**

**DETAILED EXPLANATION:**

- Capture repeating patterns
- Weekly, monthly, quarterly effects
- Example: Retail sales higher in December
- Control for seasonal variation

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** They capture, not remove, seasonality
- **C:** This is a different concept
- **D:** Trends are different from seasonality

**EXAM TRAP:** Seasonal dummies = capture repeating patterns!

**QUESTION:** Why might you apply a logarithmic transformation to variables?

**OPTIONS:**

- ❶ To make the model non-linear
- ❷ To handle different scales or non-linear relationships
- ❸ To remove all variation
- ❹ To make variables categorical

**ANSWER: B**

**DETAILED EXPLANATION:**

- Handles different scales
- Can address heteroskedasticity
- Changes interpretation of coefficients
- Model remains linear in parameters

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Model is still linear in parameters
- **C:** Variation remains, but transformed
- **D:** Variables remain continuous

**EXAM TRAP:** Log transformations = **handle different scales and non-linearity!**

**QUESTION:** What does R-squared measure in regression?

**OPTIONS:**

- ❶ The proportion of variance explained by the model
- ❷ The significance of coefficients
- ❸ The correlation between features
- ❹ The error term

**ANSWER: A**

**DETAILED EXPLANATION:**

- Ranges from 0 to 1
- Higher = better fit
- Proportion of total variation explained
- $R^2 = 1 - \frac{SSE}{SST}$

**WHY OTHER OPTIONS ARE WRONG:**

- **B:** Significance is p-values
- **C:** Correlation is different
- **D:** Error term is separate

**EXAM TRAP:** R-squared = variance explained by model!

**QUESTION:** Why use adjusted R-squared instead of R-squared?

**OPTIONS:**

- ❶ It is always larger than R-squared
- ❷ It penalizes model complexity (number of features)
- ❸ It measures significance
- ❹ It is easier to calculate

**ANSWER: B**

**DETAILED EXPLANATION:**

- R-squared always increases with more features
- Adjusted R-squared penalizes adding features
- Formula:  $\bar{R}^2 = 1 - \frac{(1-R^2)(n-1)}{n-k-1}$
- Helps prevent overfitting

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Adjusted R-squared is usually smaller
- **C:** Significance is p-values
- **D:** R-squared is easier to calculate

**EXAM TRAP:** Adjusted R-squared = penalizes model complexity!

**QUESTION: What is overfitting in regression?**

**OPTIONS:**

- ❶ Model fits training data perfectly
- ❷ Model captures noise instead of true relationship
- ❸ Model fits test data better
- ❹ Model has too few features

**ANSWER: B**

**DETAILED EXPLANATION:**

- Too complex model
- Fits noise in training data
- Poor performance on new data
- Caused by too many features or high-order terms

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Overfitting fits training perfectly but poorly generalizes
- **C:** Overfitting fails on test data
- **D:** Overfitting has too many features

**EXAM TRAP:** Overfitting = capturing noise, not signal!

**QUESTION: What is underfitting in regression?**

**OPTIONS:**

- ❶ Model is too complex
- ❷ Model is too simple to capture relationships
- ❸ Model fits training data perfectly
- ❹ Model has too many features

**ANSWER: B**

**DETAILED EXPLANATION:**

- Model is too simple
- Cannot capture true relationship
- High bias, low variance
- Example: Linear model for quadratic data

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** This is overfitting
- **C:** This is overfitting
- **D:** This is overfitting

**EXAM TRAP:** Underfitting = model too simple!

**QUESTION:** What is the bias-variance tradeoff?

**OPTIONS:**

- ❶ Tradeoff between model complexity and interpretability
- ❷ Tradeoff between model simplicity (bias) and sensitivity to data (variance)
- ❸ Tradeoff between training and test error
- ❹ Tradeoff between features and observations

**ANSWER: B**

**DETAILED EXPLANATION:**

- Simple models: high bias, low variance
- Complex models: low bias, high variance
- Need to balance for optimal performance
- Central to machine learning

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** This is accuracy-interpretability tradeoff
- **C:** This is a consequence, not definition
- **D:** This is different concept

**EXAM TRAP:** Bias-variance = **simplicity vs complexity!**

**QUESTION:** What is regularization in regression?

**OPTIONS:**

- ❶ Adding penalty terms to reduce overfitting
- ❷ Removing all features
- ❸ Increasing model complexity
- ❹ Ignoring errors

**ANSWER: A**

**DETAILED EXPLANATION:**

- Adds penalty on coefficients
- Reduces model complexity
- Helps prevent overfitting
- Examples: LASSO, Ridge (Chapter 7)

**WHY OTHER OPTIONS ARE WRONG:**

- **B:** Regularization shrinks, doesn't remove all
- **C:** Regularization REDUCES complexity
- **D:** Regularization penalizes errors

**EXAM TRAP:** Regularization = **penalty to reduce overfitting!**

**QUESTION:** What does AIC measure in stepwise regression?

**OPTIONS:**

- ❶ Model accuracy
- ❷ Prediction error penalized for model complexity
- ❸ Correlation between features
- ❹ Significance of coefficients

**ANSWER: B**

**DETAILED EXPLANATION:**

- AIC = Akaike Information Criterion
- Measures prediction error
- Penalizes models with more features
- Lower AIC indicates better model

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** AIC is not directly accuracy
- **C:** Correlation is separate
- **D:** Significance is p-values

**EXAM TRAP:** AIC = prediction error + complexity penalty!

**QUESTION:** In forward selection, when does the process stop?

**OPTIONS:**

- ❶ When all features are included
- ❷ When adding any feature no longer improves the criterion
- ❸ After exactly 3 steps
- ❹ When AIC reaches zero

**ANSWER: B**

**DETAILED EXPLANATION:**

- Start with intercept only
- Add features one by one
- Stop when improvement stops
- Or when AIC starts increasing

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** May stop before all features
- **C:** Not a fixed number of steps
- **D:** AIC is never zero in practice

**EXAM TRAP:** Forward selection stops when **no improvement in criterion!**

# Q73: Backward Selection Process - Tutorial

**QUESTION:** In backward selection, when does the process stop?

**OPTIONS:**

- ❶ When all features are removed
- ❷ When removing any feature no longer improves the criterion
- ❸ After exactly 3 steps
- ❹ When AIC reaches zero

**ANSWER: B**

**DETAILED EXPLANATION:**

- Start with all features
- Remove features one by one
- Stop when removal worsens criterion
- Or when AIC starts increasing

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** May stop before all removed
- **C:** Not a fixed number of steps
- **D:** AIC is never zero in practice

**EXAM TRAP:** Backward selection stops when **removal no longer improves!**

**QUESTION:** What is a limitation of stepwise regression?

**OPTIONS:**

- ❶ It always finds the best model
- ❷ It may not find the absolute best model and explores limited paths
- ❸ It is computationally impossible
- ❹ It requires no domain knowledge

**ANSWER: B**

**DETAILED EXPLANATION:**

- Heuristic approach
- May miss optimal combinations
- Forward and backward may give different results
- Not guaranteed to find absolute best subset

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** It may NOT find the best model
- **C:** It is computationally feasible
- **D:** Domain knowledge is still needed

**EXAM TRAP:** Stepwise is **heuristic**, not guaranteed optimal!

**QUESTION:** Why is best subset selection impractical for many features?

**OPTIONS:**

- ❶ Because it's too simple
- ❷ Because there are  $2^m$  possible models to evaluate
- ❸ Because it uses no criteria
- ❹ Because it requires no data

**ANSWER: B**

**DETAILED EXPLANATION:**

- For  $m$  features,  $2^m$  possible models
- With  $m=20$ , over 1 million models
- Computationally expensive
- Stepwise is a practical alternative

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** It is computationally complex
- **C:** It uses criteria like AIC
- **D:** It requires data

**EXAM TRAP:** Best subset has  $2^m$  models to evaluate!

**QUESTION:** Which is NOT a binary outcome in risk management?

**OPTIONS:**

- ❶ Loan defaults
- ❷ Fraud detection
- ❸ Employee salary
- ❹ Equipment failure

**ANSWER: C**

**DETAILED EXPLANATION:**

- Employee salary is continuous
- Default, fraud, failure are binary
- Binary outcomes are Yes/No
- Salary is a numerical value

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Default = Yes/No
- **B:** Fraud = Yes/No
- **D:** Failure = Yes/No

**EXAM TRAP:** Salary is **continuous**, not binary!

**QUESTION:** What is a common application of logistic regression?

**OPTIONS:**

- ❶ Predicting house prices
- ❷ Credit risk assessment (default prediction)
- ❸ Stock return forecasting
- ❹ Estimating company revenue

**ANSWER: B**

**DETAILED EXPLANATION:**

- Logistic regression for binary outcomes
- Default prediction is binary (default/no default)
- Credit risk is common application
- Example: LendingClub case

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** House prices are continuous (regression)
- **C:** Stock returns are continuous (regression)
- **D:** Revenue is continuous (regression)

**EXAM TRAP:** Logistic regression = **binary classification** like default prediction!

**QUESTION: What is nominal data?**

**OPTIONS:**

- ❶ Data with natural ordering
- ❷ Data with categories but no ordering
- ❸ Continuous numerical data
- ❹ Binary data

**ANSWER: B**

**DETAILED EXPLANATION:**

- Categories have no inherent ranking
- Example: Transport modes (car, bus, bicycle)
- Multinomial logit is appropriate
- Different from ordinal data

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** This is ordinal data
- **C:** This is continuous data
- **D:** Binary is a special case

**EXAM TRAP:** Nominal = categories without ordering!

**QUESTION:** Which is an example of ordinal data?

**OPTIONS:**

- ❶ Customer satisfaction (Very satisfied, Satisfied, Neutral, Unsatisfied)
- ❷ Color of cars
- ❸ City of residence
- ❹ Employee ID number

**ANSWER: A**

**DETAILED EXPLANATION:**

- Has natural ordering
- Very satisfied  $\prec$  Satisfied  $\prec$  Neutral  $\prec$  Unsatisfied
- Ordered logit is appropriate
- Other examples: credit ratings, risk scores

**WHY OTHER OPTIONS ARE WRONG:**

- **B:** Colors have no ordering (nominal)
- **C:** Cities have no ordering (nominal)
- **D:** IDs have no ordering (nominal)

**EXAM TRAP:** Ordinal = ordered categories!

**QUESTION:** How are prior probabilities estimated in LDA?

**OPTIONS:**

- ❶ Always set to 0.5
- ❷ Estimated from class frequencies in training data
- ❸ Randomly assigned
- ❹ Set to 1

**ANSWER: B**

**DETAILED EXPLANATION:**

- $\pi_j = \frac{n_j}{N}$
- $n_j$ : number of training observations in class  $j$
- $N$ : total training observations
- Can also be assumed equal if desired

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Not always 0.5
- **C:** Not random
- **D:** Not always 1

**EXAM TRAP:** Priors = class frequencies in training data!

**QUESTION:** In LDA, what determines the classification?

**OPTIONS:**

- ❶ The lowest discriminant score
- ❷ The highest discriminant score
- ❸ The mean of all scores
- ❹ Random selection

**ANSWER: B**

**DETAILED EXPLANATION:**

- Calculate  $\delta_j(x)$  for each class
- Choose class with maximum  $\delta_j(x)$
- This maximizes posterior probability
- Linear in features

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Lowest score would be worst
- **C:** Mean doesn't determine classification
- **D:** Classification is deterministic

**EXAM TRAP:** LDA chooses **highest discriminant score!**

# Q82: Multinomial Logit vs Ordered Logit - Tutorial

**QUESTION:** When would you use ordered logit instead of multinomial logit?

**OPTIONS:**

- ① When categories have natural ordering
- ② When categories are unordered
- ③ When predicting continuous values
- ④ When there are only two categories

**ANSWER: A**

**DETAILED EXPLANATION:**

- Ordered logit for ordinal data
- Multinomial logit for nominal data
- Ordered logit preserves ordering information
- Example: Credit ratings (AAA, AA, A, BBB)

**WHY OTHER OPTIONS ARE WRONG:**

- **B:** Multinomial is for unordered
- **C:** Neither is for continuous
- **D:** Binary logistic for two categories

**EXAM TRAP:** Ordered logit = ordered categories!

## Q83: LDA Example Calculation - Tutorial

**QUESTION:** In the LDA example, what is the inverse of the covariance matrix?

**OPTIONS:**

1  $\begin{pmatrix} 1 & 0.58 \\ 0.58 & 1 \end{pmatrix}$

2  $\begin{pmatrix} 1.52 & 0.88 \\ 0.88 & 1.52 \end{pmatrix}$

3  $\begin{pmatrix} -1.52 & -0.88 \\ -0.88 & -1.52 \end{pmatrix}$

4  $\begin{pmatrix} 0.88 & 1.52 \\ 1.52 & 0.88 \end{pmatrix}$

**ANSWER: B**

**DETAILED EXPLANATION:**

- Original covariance:  $\begin{pmatrix} 1 & -0.58 \\ -0.58 & 1 \end{pmatrix}$
- Inverse:  $\begin{pmatrix} 1.52 & 0.88 \\ 0.88 & 1.52 \end{pmatrix}$
- Used in discriminant function
- Features are standardized (variance = 1)

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** This is the covariance matrix
- **C:** Wrong signs
- **D:** Wrong arrangement

**EXAM TRAP:** Inverse covariance matrix =  $\begin{pmatrix} 1.52 & 0.88 \\ 0.88 & 1.52 \end{pmatrix}$ !

**QUESTION:** In the LDA example, why is the new borrower classified as no default?

**OPTIONS:**

- 1 Because the default score is higher
- 2 Because the no-default score is higher ( $0.30 > -3.95$ )
- 3 Because of random chance
- 4 Because there are more non-defaults in training

**ANSWER: B**

**DETAILED EXPLANATION:**

- $\delta_0 = 0.30$  (no default)
- $\delta_1 = -3.95$  (default)
- $0.30 > -3.95$ , so classify as no default
- Discriminant score much higher for no default

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Default score is lower
- **C:** Classification is deterministic
- **D:** Priors help but scores determine

**EXAM TRAP:** Higher discriminant score = preferred classification!

**QUESTION: When is LDA preferred over logistic regression?**

**OPTIONS:**

- ❶ For binary classification only
- ❷ For multi-class classification with well-separated classes
- ❸ For continuous prediction
- ❹ For clustering

**ANSWER: B**

**DETAILED EXPLANATION:**

- LDA handles multiple classes naturally
- Works well when classes are well-separated
- Logistic regression needs extension for multi-class
- LDA is robust even when assumptions violated

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Logistic regression is for binary
- **C:** Neither is for continuous
- **D:** Neither is for clustering

**EXAM TRAP:** LDA = multi-class with well-separated classes!

**QUESTION:** What are two methods to detect outliers in regression?

**OPTIONS:**

- ① Examine residuals and Cook's distance
- ② Examine R-squared and p-values
- ③ Examine correlation and covariance
- ④ Examine intercept and slope

**ANSWER: A**

**DETAILED EXPLANATION:**

- Residual plots: look for large residuals
- Cook's distance: measures influence of each point
- Large Cook's distance warrants investigation
- Both help identify influential points

**WHY OTHER OPTIONS ARE WRONG:**

- **B:** R-squared and p-values don't detect outliers
- **C:** Correlation doesn't detect outliers
- **D:** Intercept/slope don't detect outliers

**EXAM TRAP:** Outliers detected by **residuals** and **Cook's distance**!

**QUESTION:** How does stepwise regression work?

**OPTIONS:**

- ❶ It randomly selects features
- ❷ It adds or removes features based on a criterion
- ❸ It always includes all features
- ❹ It never removes features

**ANSWER: B**

**DETAILED EXPLANATION:**

- Forward: start with no features, add best one each step
- Backward: start with all features, remove worst one each step
- Uses AIC or similar criterion
- Stops when no improvement

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Stepwise is systematic, not random
- **C:** Stepwise selects subset, not all
- **D:** Stepwise can remove features

**EXAM TRAP:** Stepwise = systematic add/remove based on criterion!

**QUESTION:** Why can't linear regression be used for binary outcomes?

**OPTIONS:**

- ❶ Because it's too complex
- ❷ Because predictions can fall outside 0-1 range
- ❸ Because it always gives 0.5
- ❹ Because it requires categorical features

**ANSWER: B**

**DETAILED EXPLANATION:**

- Linear regression gives unbounded predictions
- Could predict negative probabilities
- Could predict probabilities  $\hat{p} > 1$
- Logistic regression bounds output between 0 and 1

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Linear regression is simpler
- **C:** Linear regression doesn't always give 0.5
- **D:** Linear regression doesn't require categorical features

**EXAM TRAP:** Linear regression = **unbounded predictions** (can be outside 0-1)!

**QUESTION:** What are the main assumptions underlying LDA?

**OPTIONS:**

- ❶ Features are normally distributed with common covariance
- ❷ Features are independent and identically distributed
- ❸ Features are uniformly distributed
- ❹ Features are categorical

**ANSWER: A**

**DETAILED EXPLANATION:**

- Multivariate normal for each class
- Different mean vectors per class
- Common covariance matrix
- Features must be continuous

**WHY OTHER OPTIONS ARE WRONG:**

- **B:** Features need not be independent
- **C:** Normal, not uniform
- **D:** Features are continuous, not categorical

**EXAM TRAP:** LDA = normal distributions + common covariance!

**QUESTION:** In the ROA and SIZE example, is the relationship linear?

**OPTIONS:**

- 1 Yes, it's perfectly linear
- 2 No, because the equation includes a squared term
- 3 Yes, because all coefficients are positive
- 4 No, because ROA is in percent

**ANSWER: B**

**DETAILED EXPLANATION:**

- Equation:  $ROA = 2.6 + 4.22(SIZE) - 1.32(SIZE)^2$
- Squared term creates non-linearity
- Relationship changes with SIZE
- Maximum at  $SIZE = 1.60$

**WHY OTHER OPTIONS ARE WRONG:**

- A: Squared term makes it non-linear
- C: Positive coefficients don't imply linearity
- D: Units don't determine linearity

**EXAM TRAP:** Squared term = non-linear relationship!

# Q91: ROA Calculation Example - Tutorial

**QUESTION:** For the ROA model, what is the expected ROA for  $SIZE = 1.0$ ?

**OPTIONS:**

- ☒ 1 4.50%
- ☐ 2 5.50%
- ☐ 3 6.50%
- ☐ 4 7.50%

**ANSWER: B**

**DETAILED EXPLANATION:**

- Equation:  $ROA = 2.6 + 4.22(1.0) - 1.32(1.0)^2$
- $ROA = 2.6 + 4.22 - 1.32$
- $ROA = 5.50$
- SIZE measured in hundreds of millions

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Too low
- **C:** Too high
- **D:** Too high

**EXAM TRAP:** Plug  $SIZE = 1.0$  into equation:  $ROA = 5.50\%$ !

**QUESTION:** In the ROA example, what SIZE maximizes ROA?

**OPTIONS:**

- ☒ 1. 1.00
- ☐ 2. 1.60
- ☐ 3. 2.00
- ☐ 4. 0.50

**ANSWER: B**

**DETAILED EXPLANATION:**

- Derivative:  $\frac{dROA}{dSIZE} = 4.22 - 2.64(SIZE)$
- Set to zero:  $4.22 - 2.64(SIZE) = 0$
- $SIZE = 4.22/2.64 = 1.60$
- This is \$160 million (since SIZE is in hundreds of millions)

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Not the maximum
- **C:** Beyond the maximum
- **D:** Below the maximum

**EXAM TRAP:** Optimal SIZE = 1.60 (hundreds of millions)!

**QUESTION:** What is the maximum ROA in the example?

**OPTIONS:**

- 1 5.50%
- 2 6.03%
- 3 7.00%
- 4 8.50%

**ANSWER: B**

**DETAILED EXPLANATION:**

- Plug  $SIZE = 1.60$  into equation
- $ROA = 2.6 + 4.22(1.60) - 1.32(1.60)^2$
- $ROA = 2.6 + 6.752 - 3.3792$
- $ROA = 6.03\%$

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** This is ROA at  $SIZE = 1.0$
- **C:** Too high
- **D:** Too high

**EXAM TRAP:** Maximum ROA = 6.03% at  $SIZE = 1.60$ !

**QUESTION:** In the marketing campaign example, what are the features?

**OPTIONS:**

- ❶ Age and Income
- ❷ Balance and Age
- ❸ Balance and Income
- ❹ Age and Gender

**ANSWER: B**

**DETAILED EXPLANATION:**

- Features: Balance (dollars) and Age (years)
- Target: Subscribed (Yes/No)
- Binary classification problem
- LDA used to classify clients

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Income is not a feature
- **C:** Income is not a feature
- **D:** Gender is not a feature

**EXAM TRAP:** Features = Balance and Age!

**QUESTION:** In the marketing example, what are the prior probabilities?

**OPTIONS:**

- ❶ 0.5 and 0.5
- ❷ 0.6 and 0.4
- ❸ 0.7 and 0.3
- ❹ 0.8 and 0.2

**ANSWER: B**

**DETAILED EXPLANATION:**

- 6 out of 10 subscribed (60%)
- 4 out of 10 did not subscribe (40%)
- $\pi_1 = 0.6$  (subscribed)
- $\pi_2 = 0.4$  (not subscribed)

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Not balanced (6 vs 4)
- **C:** Not 7 vs 3
- **D:** Not 8 vs 2

**EXAM TRAP:** Priors = 0.6 and 0.4 from data!

**QUESTION:** In the marketing example, what is the mean balance for subscribers?

**OPTIONS:**

- 1 -0.44
- 2 0.29
- 3 0.00
- 4 1.00

**ANSWER: B**

**DETAILED EXPLANATION:**

- $\mu_1 = \begin{pmatrix} 0.29 \\ 0.00 \end{pmatrix}$
- First element: mean balance for subscribers
- Second element: mean age for subscribers
- Features are standardized

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** This is mean balance for non-subscribers
- **C:** This is mean age for subscribers
- **D:** Not correct

**EXAM TRAP:** Subscriber balance mean = 0.29 (standardized)!

**QUESTION:** In the marketing example, what is the mean balance for non-subscribers?

**OPTIONS:**

- 1 0.29
- 2 -0.44
- 3 0.00
- 4 -1.00

**ANSWER: B**

**DETAILED EXPLANATION:**

- $\mu_2 = \begin{pmatrix} -0.44 \\ 0.00 \end{pmatrix}$
- First element: mean balance for non-subscribers
- Second element: mean age for non-subscribers
- Non-subscribers have lower average balance

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** This is mean balance for subscribers
- **C:** This is mean age
- **D:** Not correct

**EXAM TRAP:** Non-subscriber balance mean = -0.44 (standardized)!

**QUESTION:** In the marketing example, what is the covariance between features?

**OPTIONS:**

- 1 0.33
- 2 0.58
- 3 -0.58
- 4 1.00

**ANSWER: A**

**DETAILED EXPLANATION:**

- Covariance matrix:  $\begin{pmatrix} 1 & 0.33 \\ 0.33 & 1 \end{pmatrix}$
- Variances = 1 (features standardized)
- Covariance = 0.33
- Positive correlation between balance and age

**WHY OTHER OPTIONS ARE WRONG:**

- **B:** This is from LDA example
- **C:** This is from LDA example
- **D:** This is variance, not covariance

**EXAM TRAP:** Covariance = 0.33 in marketing example!

**QUESTION:** In the marketing example, how would a 51-year-old with \$10,635 be classified?

**OPTIONS:**

- ❶ Subscribed
- ❷ Not subscribed
- ❸ Cannot be determined
- ❹ Both equally likely

**ANSWER: A**

**DETAILED EXPLANATION:**

- Standardized values: Age = 0.65, Balance = 1.64
- LDA would classify based on discriminant scores
- Higher balance and older age suggest subscription
- Subscriber class has higher mean balance

**WHY OTHER OPTIONS ARE WRONG:**

- **B:** Features suggest subscriber
- **C:** We have enough information
- **D:** Scores would not be equal

**EXAM TRAP:** Classification uses **discriminant scores** to decide!

**QUESTION:** What is the most comprehensive summary of supervised learning for numerical data?

**OPTIONS:**

- ❶ Only linear regression is used
- ❷ Models include linear regression (continuous outcomes) and logistic regression/LDA (classification)
- ❸ Only logistic regression is used
- ❹ Only LDA is used

**ANSWER: B**

**DETAILED EXPLANATION:**

- **Continuous outcomes:** Linear regression (simple, multiple)
- **Classification:** Logistic regression, LDA
- **Extensions:** Multinomial logit, ordered logit
- **Key concepts:** OLS, AIC, stepwise, assumptions

**WHY OTHER OPTIONS ARE WRONG:**

- **A:** Multiple techniques are used
- **C:** Linear regression and LDA also used
- **D:** Linear and logistic also used

**EXAM SUCCESS:** Understand regression and classification techniques!